

Appendix A. Trial-time implementation of the GPT-4o mini-program intervention

This appendix describes the intervention as implemented and tested in the present trial. It focuses on the task structure, the LungDiag-derived knowledge layer, the trial-time workflow, the operational boundaries of the evaluated intervention, and the platform safeguards directly relevant to interpretation of the study.

A1. Overview of the tested intervention

The intervention was delivered through a WeChat mini-program built on OpenAI's GPT-4o. Within the trial, GPT-4o was used only to support three pre-specified tasks during questionnaire completion: (1) identification of risk-related behaviors or etiologic clues, (2) selection of the most likely preliminary diagnosis, and (3) generation of a triage recommendation. The questionnaire itself was an independently constructed evaluation instrument and was not part of the LungDiag-derived knowledge resource.

A2. LungDiag-derived knowledge resource

The mini-program incorporated a LungDiag-derived respiratory knowledge resource through retrieval-augmented generation. In the original LungDiag framework, the knowledge resource was developed from multicenter electronic health records and was designed to support respiratory disease classification through natural language processing, clinical feature extraction, and standardized phenotypic representation. Its clinical terminology resources covered disease names, symptoms, laboratory tests, imaging findings, medications, and surgical procedures. Within the original LungDiag framework, respiratory conditions were standardized into 10 major disease categories.

A3. Separation between the knowledge layer and the questionnaire

In the present trial, the LungDiag-derived resource served as an evidence support layer rather than as a case bank. The questionnaire and case materials used for outcome assessment were constructed separately as study-specific evaluation instruments. Accordingly, the intervention tested in this trial should be understood as the combination of GPT-4o, a LungDiag-derived respiratory knowledge layer, and a structured case-based assessment process, rather than as direct use of the knowledge resource itself as the source of the trial questions.

A4. Trial-time task workflow

1. Participant-entered symptom- or case-related information was submitted through the WeChat mini-program.
2. The system retrieved relevant respiratory content from the LungDiag-derived knowledge resource.
3. The retrieved content was paired with task-specific prompts corresponding to the target task.
4. GPT-4o generated an evidence-grounded output for one of the three predefined tasks.
5. Retrieved content was limited to respiratory-domain material within the LungDiag-derived knowledge resource, and outputs were constrained to the three structured task types used in trial scoring.

A5. Task-specific prompting framework

To support consistent task-oriented outputs, retrieved respiratory content was paired with task-specific prompts. These prompts were structured according to the three pre-specified trial tasks rather than as unrestricted open-ended consultation prompts.

Template	Input focus	Output goal
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Template 1. Risk-related behavior or etiologic clue identification	Symptom descriptions, exposures, behaviors, or contextual clues relevant to respiratory disease.	Identify the most relevant risk-related behavior(s) or etiologic clue(s) supported by the presented information.
Template 2. Preliminary diagnosis selection	Chief complaint, symptom pattern, and related clinical clues.	Select the most likely preliminary diagnosis from the information provided in the case.
Template 3. Triage recommendation	Symptom severity, warning signs, and case context.	Generate a triage recommendation aligned with the structured task design used in the trial.

A6. Platform safeguards, operational boundaries, and reporting scope

The platform was deployed on Tier-III-classified servers with end-to-end encryption. Its operation was monitored through automated logging and rule-based safeguards designed to maintain use within the intended respiratory disease-related task boundaries of the trial. These safeguards were relevant to the structured questionnaire workflow and were not intended to support unrestricted open-ended consultation outside the study tasks.

This appendix is intended to clarify the trial-time intervention that was actually tested. It therefore focuses on the task structure, the LungDiag-derived knowledge support layer, the retrieval-linked workflow, and the platform safeguards described in the revised manuscript.

A7. Trial-time deployment boundaries and enabled functions

The intervention evaluated in this trial used a fixed task structure throughout participant enrollment. The tested workflow was limited to text-based questionnaire completion within the WeChat mini-program and the three pre-specified outputs described above: identification of risk-related behaviors or etiologic clues, selection of the most likely preliminary diagnosis, and generation of a triage recommendation. No unrestricted open-ended consultation or external web search was included in the evaluated workflow. Functions not prospectively incorporated into the structured questionnaire workflow and study scoring process were not considered part of the evaluated intervention. If additional features were available in the broader product environment outside the trial setting, they were not part of the present evaluation.

A8. Availability and reproducibility statement

Code to reproduce the reported statistical results is available in the repository listed in the Data availability statement of the manuscript. Public non-proprietary supplementary materials in that repository include the present appendix, Supplementary Appendix B (scoring rubric and questionnaire administration), Supplementary Appendix C (composition of the 88-case respiratory question bank), and Supplementary Data 1 (triage urgency source data). Reviewer-only supplementary materials provided with the revised submission include the questionnaire materials and answer keys. The trial-specific WeChat mini-program code, internal prompt templates, and the LungDiag-derived knowledge resource are not further disclosed in this report.